

SQL FOR DATA ANALYSIS

Objective: Use SQL queries to extract and analyze data from a database.

Tools: MySQL

Basic Analysis

1. How many users are registered in the database?

select distinct count(user_id) as 'Total Users' from users;

```
MariaDB [ecommerce]> select distinct count(user_id) as 'Total Users' from users;
+-----+
| Total Users |
+-----+
|         10000 |
+-----+
1 row in set (0.70 sec)
```

2. How many orders have been placed?

select distinct count(order_id) as 'Total Orders' from orders;

```
MariaDB [ecommerce]> select distinct count(order_id) as 'Total Orders' from orders;
+-----+
| Total Orders |
+-----+
|         20000 |
+-----+
1 row in set (1.99 sec)
```

3. Find count of all orders with a specific status (e.g., Delivered, Cancelled).

select distinct order_status, count(order_status) as "No of Orders" from orders group by order_status;

```
MariaDB [ecommerce]> select distinct order_status, count(order_status) as "No of Orders" from orders group by order_status;
+-----+-----+
| order_status | No of Orders |
+-----+-----+
| cancelled    |          3920 |
| completed    |          4021 |
| processing    |          3880 |
| returned     |          4066 |
| shipped       |          4113 |
+-----+-----+
5 rows in set (0.28 sec)
```

4. Find count of all orders placed in 2024.

select count(order_id) as 'Order Count' from orders where Year(order_date) = 2024;

```
MariaDB [ecommerce]> select count(order_id) as 'Order Count' from orders where Year(order_date) = 2024;
+-----+
| Order Count |
+-----+
|         10769 |
+-----+
1 row in set (0.04 sec)
```

5. Find count of all orders in all months.

select MONTHNAME(order_date) as 'Month', count(order_id) as 'Order_Count' from orders group by Month order by Order_Count;

```
MariaDB [ecommerce]> select MONTHNAME(order_date) as 'Month', count(order_id) as 'Order_Count' from orders group by Month order by Order_Count;
```

Month	Order_Count
December	888
November	1264
February	1639
June	1765
January	1767
September	1788
April	1791
October	1792
August	1807
May	1810
March	1835
July	1854

```
2 rows in set (0.05 sec)
```

Customer Analysis

6. Which city has the highest number of users?

select city, count(user_id) as 'Total_Users' from users order by Total_users desc limit 1;

```
MariaDB [ecommerce]> select city, count(user_id) as 'Total_Users' from users order by Total_users desc limit 1;
```

city	Total_Users
New Roberttown	10000

```
1 row in set (0.33 sec)
```

7. Find gender based order count.

select gender, count(user_id) as Total from users group by gender;

```
MariaDB [ecommerce]> select gender, count(user_id) as Total from users group by gender;
```

gender	Total
Female	3334
Male	3247
Other	3419

```
3 rows in set (0.07 sec)
```

8. Who are the top 10 customers based on the number of orders?

select u.name as 'Name', count(o.order_id) as 'Total_Orders' from users u join orders o on u.user_id = o.user_id group by u.user_id, u.name order by Total_Orders desc limit 10;

```
MariaDB [ecommerce]> select u.name as 'Name', count(o.order_id) as 'Total_Orders' from users u
-> join orders o on u.user_id = o.user_id group by u.user_id, u.name order by Total_Orders desc limit 10;
```

Name	Total_Orders
Eric Miller	10
Daniel Martinez	9
Robert Jennings	9
Emily Hayes	8
Shawn Rodriguez	8
Donna Morse	8
Glenn Duffy	8
Melissa Allen	8
Michael Horton	8
Robert Smith	8

```
10 rows in set (28.23 sec)
```

9. Which customers have never placed an order?

select name as 'Name' from users where user_id not in (select user_id from orders) group by Name;

Troy Spencer
Tyler Anderson
Tyler Gonzalez
Tyler Hogan
Tyler Munoz
Tyler Navarro
Tyler Palmer
Tyler Walker
Valerie Powell

10. Top 10 average number of orders per customer?

select count(o.order_id) / count(distinct u.user_id) as 'Avg_order_per_customers' from orders o join users u on u.user_id = o.user_id;

```
MariaDB [ecommerce]> select count(o.order_id) / count(distinct u.user_id) as 'Avg_order_per_customers' from orders o join
-> on u.user_id = o.user_id;
+-----+
| Avg_order_per_customers |
+-----+
| 2.3162 |
+-----+
1 row in set, 1 warning (22.25 sec)
```

Sales Analysis

11. What is the total revenue generated?

select sum(total_amount) as Revenue from orders;

```
MariaDB [ecommerce]> select sum(total_amount) as Revenue from orders;
+-----+
| Revenue |
+-----+
| 11918668.95 |
+-----+
1 row in set (0.07 sec)
```

12. What is the average order value?

select avg(total_amount) as 'Avg_order' from orders;

```
MariaDB [ecommerce]> select avg(total_amount) as 'Avg_order' from orders;
+-----+
| Avg_order |
+-----+
| 595.933448 |
+-----+
1 row in set (0.02 sec)
```

13. Which month generated the highest sales?

select monthname(order_date) as 'Month',sum(total_amount) as Revenue from orders group by Month order by Revenue desc;

```
MariaDB [ecommerce]> select monthname(order_date) as 'Month',sum(total_amount) as Revenue from orders group by Month order by Revenue desc;;
+-----+-----+
| Month | Revenue |
+-----+-----+
| July | 1137336.74 |
| May | 1091372.77 |
| April | 1077469.41 |
| June | 1056448.28 |
| September | 1051443.49 |
| October | 1050449.94 |
| March | 1046877.95 |
| January | 1046038.87 |
| August | 1031340.01 |
| February | 974090.97 |
| November | 789393.85 |
| December | 566301.65 |
+-----+-----+
12 rows in set (0.06 sec)
```

14. Which day had the highest number of orders?

select DAYNAME(order_date) as Day, sum(total_amount) as Revenue from orders group by Day order by Revenue limit 1;

```
MariaDB [ecommerce]> select DAYNAME(order_date) as Day, sum(total_amount) as Revenue from orders group by Day order by Revenue limit 1;
+-----+-----+
| Day | Revenue |
+-----+-----+
| Monday | 1612959.30 |
+-----+-----+
1 row in set (0.45 sec)
```

15. Which orders have values greater than the average order value?

select * from orders where total_amount > (select avg(total_amount) from orders) limit 1;

```
MariaDB [ecommerce]> select * from orders where total_amount > (select avg(total_amount) from orders) limit 1;
+-----+-----+-----+-----+-----+
| order_id | user_id | order_date | order_status | total_amount |
+-----+-----+-----+-----+-----+
| 000000001 | 0009310 | 2025-09-09 14:52:37 | processing | 689.66 |
+-----+-----+-----+-----+-----+
1 row in set (0.03 sec)
```

Product Analysis

16. Which products are sold the most?

select p.product_name as 'Most_Sold_product', sum(oi.quantity) as 'Total_Sold' from products p inner join order_items oi on p.product_id = oi.product_id group by p.product_name order by Total_Sold desc limit 1;

```
MariaDB [ecommerce]> select p.product_name as 'Most_Sold_product', sum(oi.quantity) as 'Total_Sold' from products p inner join order_items oi
-> on p.product_id = oi.product_id group by p.product_name order by Total_Sold desc limit 1;
+-----+-----+
| Most_Sold_product | Total_Sold |
+-----+-----+
| Solace Particular | 104 |
+-----+-----+
1 row in set (12.71 sec)
```

17. Which products generate the highest revenue?

select p.product_name as 'Product', sum(oi.quantity * p.price) as 'Revenue' from products p inner join order_items oi on p.product_id = oi.product_id group by Product order by Revenue desc limit 1;

```
MariaDB [ecommerce]> select p.product_name as 'Product', sum(oi.quantity * p.price) as 'Revenue' from products p
-> inner join order_items oi on p.product_id = oi.product_id group by Product order by Revenue desc limit 1;
+-----+-----+
| Product | Revenue |
+-----+-----+
| Willow Result | 73525.32 |
+-----+-----+
1 row in set (17.12 sec)
```

18. Which product categories have the most products?

select category, count(Product_id) as 'Total_Products' from products group by category order by Total_Products desc;

```
MariaDB [ecommerce]> select category, count(Product_id) as 'Total_Products' from products group by category order by Total_Products desc;
+-----+-----+
| category | Total_Products |
+-----+-----+
| Clothing | 213 |
| Toys | 213 |
| Pet Supplies | 202 |
| Beauty | 202 |
| Electronics | 203 |
| Home & Kitchen | 199 |
| Books | 194 |
| Automotive | 173 |
| Groceries | 183 |
| Sports | 183 |
+-----+-----+
10 rows in set (0.00 sec)
```

19. Which products have never been ordered?

select product_name from products where product_id not in (select product_id from order_items);

```
MariaDB [ecommerce]> select product_name from products where product_id not in (select product_id from order_items);
Empty set (0.06 sec)
```

20. What is the average price of products?

select avg(price) as 'Avg_Price' from products;

```
MariaDB [ecommerce]> select avg(price) as 'Avg_Price' from products;
+-----+
| Avg_Price |
+-----+
| 200.342230 |
+-----+
1 row in set (0.13 sec)
```

Order Analysis

21. How many orders are in each order status?

select order_status, count(order_id) as 'Total_Orders' from orders group by order_status;

```
MariaDB [ecommerce]> select order_status, count(order_id) as 'Total_Orders' from orders group by order_status;
+-----+-----+
| order_status | Total_Orders |
+-----+-----+
| cancelled    | 3920         |
| completed    | 4021         |
| processing    | 3880         |
| returned     | 4066         |
| shipped       | 4113         |
+-----+-----+
5 rows in set (0.47 sec)
```

22. Which users placed the most orders?

select u.name as Name, count(o.order_id) as 'Total_orders' from users u inner join orders o on u.user_id=o.user_id group by u.user_id order by Total_orders desc limit 1;

```
MariaDB [ecommerce]> select u.name as Name, count(o.order_id) as 'Total_orders' from users u inner join orders o on u.user_id=o.user_id group by u.user_id
-> order by Total_orders desc limit 1;
+-----+-----+
| Name      | Total_orders |
+-----+-----+
| Eric Miller | 10           |
+-----+-----+
1 row in set (34.45 sec)
```

23. What is the average number of items per order?

select sum(quantity) / count(distinct order_id) as 'AvgItemsPerOrder' from order_items;

```
MariaDB [ecommerce]> select sum(quantity) / count(distinct order_id) as 'AvgItemsPerOrder' from order_items;
+-----+
| AvgItemsPerOrder |
+-----+
| 3.0409           |
+-----+
1 row in set (0.31 sec)
```

24. Which orders contain more than five items?

select order_id from order_items group by order_id having count(order_id) > 5;

```
MariaDB [ecommerce]> select order_id from order_items group by order_id having count(order_id) > 5;
+-----+
| order_id |
+-----+
| 000000052 |
| 000000069 |
| 000000267 |
| 000000289 |
| 000000306 |
| 000000362 |
+-----+
```

Advanced Analysis

1. Find customers whose spending is above the average customer spending.

```
SELECT u.user_id, u.name, SUM(o.total_amount) AS total_spent
FROM users u INNER JOIN orders o ON u.user_id = o.user_id GROUP BY
u.user_id, u.name HAVING total_spent > ( SELECT AVG(customer_total) FROM (
SELECT SUM(total_amount) AS customer_total FROM orders GROUP BY user_id
) AS subquery ) ORDER BY total_spent DESC;
```

```
+-----+-----+-----+
| user_id | name      | total_spent |
+-----+-----+-----+
| 000000052 | Charles Barnett | 1402.65     |
| 000000069 | Tammy Escobar   | 1402.58     |
| 000000267 | Daisy Hull      | 1402.09     |
| 000000289 | James Keller    | 1401.63     |
| 000000306 | Nicholas McDonald | 1400.93     |
| 000000362 | Gavin Williams  | 1400.79     |
| 000000052 | Chris Kirby     | 1400.16     |
| 000000069 | Annette Bryant  | 1399.98     |
| 000000267 | Robert Horton   | 1399.57     |
| 000000289 | Nancy Branch    | 1398.44     |
| 000000306 | Antonio Rivera  | 1398.20     |
| 000000362 | Robert Duncan   | 1395.83     |
+-----+-----+-----+
```

2. Create a view showing customer order summaries

```
CREATE VIEW customer_order_summary AS SELECT u.user_id, u.name,
COUNT(o.order_id) AS total_orders, SUM(o.total_amount) AS total_spent,
AVG(o.total_amount) AS average_order_value FROM users u JOIN orders o ON
u.user_id = o.user_id GROUP BY u.user_id, u.name;
SELECT * FROM customer_order_summary limit 10;
```

user_id	name	total_orders	total_spent	average_order_value
U0000001	Angel Hill	1	1392.89	1392.890000
U0000002	Jesse Guzman	4	1808.15	452.037500
U0000003	Adam Shaffer	3	1153.01	384.336667
U0000004	Melanie Munoz	3	7620.84	2540.280000
U0000005	Janet Williams	1	12.66	12.660000
U0000006	Michael Santiago	6	9952.28	1658.713333
U0000008	Joseph Clark	2	1520.91	760.455000
U0000009	Anna Baker	1	1297.61	1297.610000
U0000010	Brandon Rodriguez	5	1631.68	326.336000
U0000011	Tasha Valencia	2	1050.88	525.440000

10 rows in set (35.49 sec)

3. Create an Index for searching.

```
CREATE INDEX idx_user_id ON orders(user_id);
CREATE INDEX idx_order_id ON order_items(order_id);
```

4. Find the Top-Selling Product in Each Category

```
SELECT p.category, p.product_name, SUM(oi.quantity) AS total_quantity FROM
products p JOIN order_items oi ON p.product_id = oi.product_id GROUP BY
p.category, p.product_name ORDER BY p.category, total_quantity DESC;
```

Sports	Astra Dream	30
Sports	GreenLeaf Study	30
Sports	Nimbus Cell	30
Sports	Zenith Wait	30
Sports	Nimbus Significant	30
Sports	Solace Policy	29
Sports	Everest Table	29
Sports	Everest Side	29
Sports	Nimbus Threat	29
Sports	Everest Century	29
Sports	GreenLeaf Image	29
Sports	Harbor Significant	29
Sports	Everest Compare	29
Sports	GreenLeaf More	29
Sports	Everest Stock	29
Sports	Everest Child	29
Sports	Willow Back	28
Sports	Zenith Guy	28
Sports	Harbor Decade	28
Sports	Zenith Hope	28
Sports	Willow Condition	28
Sports	Acme Radio	28